

SECR2043 OPERATING SYSTEMS

Name : Iman Abadi Bin Mohd Nizwan
Student ID : A23CS0084
Section : 02

Marks

This lab assessment is designed to test your understanding and skills on some basic concepts and tools related to operating system. Please follow the instructions carefully and submit your answers in this word document and rename the file as **os-lab-assessment01-studentname-matricno.docx**.

ACTIVITY 1 : BASIC LINUX COMMANDS

In this activity, you will practice some basic Linux commands that are useful for operating system tasks. For each question, write down the command you used and the output you got in your answer file. If the command does not produce any output, write "**No output**" instead.

1.	Display your current working directory.	
	Command	Output
	pwd	<pre>iman@secr2043:~\$ pwd /home/iman</pre>
2.	List all the files and directories in your home directory, including hidden ones.	
	Command	Output
	ls -a	<pre>iman@secr2043:~\$ ls -a . .bash_history .bashrc .sudo_as_admin_success .. .bash_logout .profile example.txt</pre>
3.	Create a new directory named "os_lab" in your home directory.	
	Command	Output
	mkdir os_lab	<pre>iman@secr2043:~\$ mkdir os_lab</pre>
4.	Change your current working directory to "os_lab".	

	Command	Output
	cd os_lab pwd	<pre> iman@secr2043:~\$ cd os_lab iman@secr2043:~/os_lab\$ pwd /home/iman/os_lab </pre>
5.	Create a new file named "hello.txt" in "os_lab" and write "Hello, world!" in it.	
	touch hello.txt echo "Hello World" > hello.txt	<pre> iman@secr2043:~\$ cd os_lab iman@secr2043:~/os_lab\$ touch hello.txt iman@secr2043:~/os_lab\$ ls hello.txt iman@secr2043:~/os_lab\$ echo "Hello, world" > hello.txt </pre>
6.	Display the content of "hello.txt".	
	cat hello.txt	<pre> iman@secr2043:~/os_lab\$ cat hello.txt Hello, world </pre>
7.	Copy "hello.txt" to another file named "hello_copy.txt" in the same directory.	
	cp hello.txt hello_copy.txt	<pre> iman@secr2043:~/os_lab\$ cp hello.txt hello_copy.txt iman@secr2043:~/os_lab\$ ls hello.txt hello_copy.txt iman@secr2043:~/os_lab\$ cat hello_copy.txt Hello, world </pre>
8.	Rename "hello_copy.txt" to "hello_bak.txt".	
	mv hello_copy.txt hello_bak.txt	<pre> iman@secr2043:~/os_lab\$ mv hello_copy.txt hello_bak.txt iman@secr2043:~/os_lab\$ ls hello.txt hello_bak.txt </pre>
9.	Delete "hello.txt".	
	rm hello.txt	

		<pre> iman@secr2043:~/os_lab\$ rm hello.txt iman@secr2043:~/os_lab\$ ls hello_bak.txt </pre>
10.	Display your home directory with additional information, together with subdirectories contents.	
	Command	Output
	ls ~ -Ralh	<pre> iman@secr2043:~/os_lab\$ ls ~ -Ralh /home/iman: total 32K drwxr-x--- 4 iman iman 4.0K Apr 22 07:18 . drwxr-xr-x 6 root root 4.0K Apr 20 08:39 .. -rw----- 1 iman iman 115 Apr 21 14:26 .bash_history -rw-r--r-- 1 iman iman 220 Apr 20 08:39 .bash_logout -rw-r--r-- 1 iman iman 3.7K Apr 20 08:39 .bashrc -rw-r--r-- 1 iman iman 807 Apr 20 08:39 .profile -rw-r--r-- 1 iman iman 0 Apr 20 08:51 .sudo_as_admin_successful -rw-r--r-- 1 iman iman 0 Apr 22 06:58 example.txt drwxr-xr-x 2 iman iman 4.0K Apr 22 07:38 os_lab drwxr-xr-x 2 iman iman 4.0K Apr 22 06:58 share /home/iman/os_lab: total 12K drwxr-xr-x 2 iman iman 4.0K Apr 22 07:38 . drwxr-x--- 4 iman iman 4.0K Apr 22 07:18 .. -rw-r--r-- 1 iman iman 13 Apr 22 07:32 hello_bak.txt /home/iman/share: total 8.0K drwxr-xr-x 2 iman iman 4.0K Apr 22 06:58 . drwxr-x--- 4 iman iman 4.0K Apr 22 07:18 .. -rw-r--r-- 1 iman iman 0 Apr 22 06:58 textfile01.txt </pre>
	Total Mark:	

ACTIVITY 2 : Text File Manipulation

In this activity, you will practice some commands to create and manipulate text files using echo, cat and touch.

For each question, write down the command and the output (if any) in the answer sheet.

1.	Create an empty file named "hello.txt" in the current working directory.	
	Command	Output
	touch hello.txt	No output <pre> iman@secr2043:~/os_lab\$ touch hello.txt iman@secr2043:~/os_lab\$ ls hello.txt hello_bak.txt </pre>
2.	Write the text "Hello World" to "hello.txt" using echo.	
	Command	Output

	echo "Hello World" > hello.txt	<pre> iman@secr2043:~/os_lab\$ echo Hello World > hello.txt iman@secr2043:~/os_lab\$ cat hello.txt Hello World </pre>
3.	Append the text "This is a test" to "hello.txt" using echo.	
	Command	Output
	echo "This is a test" >> hello.txt	<pre> iman@secr2043:~/os_lab\$ echo "This is a test" >> hello.txt iman@secr2043:~/os_lab\$ cat hello.txt Hello World This is a test </pre>
4.	Display the contents of "hello.txt" using cat.	
	Command	Output
	cat hello.txt	<pre> iman@secr2043:~/os_lab\$ echo "This is a test" >> hello.txt iman@secr2043:~/os_lab\$ cat hello.txt Hello World This is a test </pre>
5.	Create another file named "bye.txt" with the text "Goodbye World" using echo.	
	Command	Output
	echo "Goodbye World" > bye.txt	<pre> iman@secr2043:~/os_lab\$ echo "Goodbye World" > bye.txt iman@secr2043:~/os_lab\$ cat bye.txt Goodbye World </pre>
6.	Concatenate the contents of "hello.txt" and "bye.txt" and display them using cat.	
	Command	Output
	cat hello.txt bye.txt	<pre> iman@secr2043:~/os_lab\$ cat hello.txt bye.txt Hello World This is a test Goodbye World </pre>
7.	Concatenate the contents of "hello.txt" and "bye.txt" and write them to a new file named "greeting.txt" using cat.	
	Command	Output
	cat hello.txt bye.txt > greetings.txt	<pre> iman@secr2043:~/os_lab\$ cat hello.txt bye.txt > greetings.txt iman@secr2043:~/os_lab\$ cat greetings.txt Hello World This is a test Goodbye World </pre>

8.	Display the contents of "greeting.txt" using cat.	
	Command	Output
	cat greetings.txt	<pre> iman@secr2043:~/os_lab\$ cat hello.txt bye.txt > greetings.txt iman@secr2043:~/os_lab\$ cat greetings.txt Hello World This is a test Goodbye World </pre>
9.	Edit the file "greeting.txt" using Nano and change the word "World" to your name in both lines.	
	Command	Output
	nano greetings.txt	No output
10.	Display the modified contents of "greeting.txt" using cat.	
	Command	Output
	cat greetings.txt	<pre> iman@secr2043:~/os_lab\$ cat greetings.txt Hello Iman This is a test Goodbye Iman </pre>
	Total Mark:	

ACTIVITY 3 : WRITE, COMPILE AND RUN C PROGRAM

In this activity, you will practice how to write, compile and run a simple C program using gcc. For each question, write down the command and the output (if any) in the answer sheet.

1.	Write a C program that prints "Hello World" to the standard output using Nano and save it as "hello.c".	
	Command	Output
	nano hello.c	<pre>iman@secr2043:~/os_lab\$ nano hello.c iman@secr2043:~/os_lab\$ gcc hello.c -o hello iman@secr2043:~/os_lab\$./hello Hello, World!</pre>
2.	Compile the program "hello.c" using gcc and generate an executable file named "hello".	
	Command	Output
	gcc hello.c -o hello	<pre>iman@secr2043:~/os_lab\$ nano hello.c iman@secr2043:~/os_lab\$ gcc hello.c -o hello iman@secr2043:~/os_lab\$./hello Hello, World!</pre>
3.	Run the executable file "hello" and display its output.	
	Command	Output
	./hello	<pre>iman@secr2043:~/os_lab\$ nano hello.c iman@secr2043:~/os_lab\$ gcc hello.c -o hello iman@secr2043:~/os_lab\$./hello Hello, World!</pre>
4.	Write a C program that takes two integers as command line arguments and prints their sum to the standard output using Nano and save it as "sum.c".	
	Command	Output
	nano sum.c	No output
5.	Compile the program "sum.c" using gcc and generate an executable file named "sum".	
	Command	Output
	gcc sum.c -o sum	No output

6.	Run the executable file "sum" with 10 and 20 as arguments and display its output.	
	Command	Output
	./sum 10 20	<pre> iman@secr2043:~/os_lab\$./sum 10 20 10 + 20 = 30 </pre>
7.	Run the executable file "sum" with -5 and 15 as arguments and display its output.	
	Command	Output
	./sum -5 15	<pre> iman@secr2043:~/os_lab\$./sum -5 15 -5 + 15 = 10 </pre>
8.	Write a C program that reads a line of text from the standard input and prints it to the standard output using Nano and save it as "echo.c".	
	Command	Output
	nano echo.c	<pre> #include <stdio.h> #include <string.h> int main() { char input[50]; // Buffer to store the name // Prompt for name printf("Enter your input: "); fgets(input, 50, stdin); // Read name from keyboard // Remove any trailing newline from the name input[strcspn(input, "\n")] = '\0'; // Display the information printf("%s\n", input); return 0; } </pre>
9.	Compile the program "echo.c" using gcc and generate an executable file named "echo".	
	Command	Output
	gcc echo.c -o echo	No Output
10.	Run the executable file "echo", type "Hello World" as input and display its output.	
	Command	Output
	./echo	<pre> iman@secr2043:~/os_lab\$./echo Enter your input: Hello World Hello World </pre>
Total Mark:		

